

Lost Items Team

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We are submitting our report entitled “Lost Item Finding Service: Final Report” In this report, we describe our observations, interviews, methods, findings and implications for the design of a technology aimed to assist users in finding commonly lost or misplaced possessions.

From the observation study in our report, major implications for the design of our service are as follows:

- A solution should leverage the spatial orientation of a user's space: participants shared a systematic search strategy, starting from an item's "home" location, then retracing steps, and only then expanding the search if necessary
- A solution should support the user emotionally: participants were consistently observed to be highly emotionally charged while searching for lost items.

After interviewing potential users of this service, we created the following major design implications:

- Users desire devices that deal with location data to be safe and secure.
- Slim profiles and a lightweight form factor are important for users for any device or peripheral that would potentially be affixed, attached, or adhered to their day to day devices.
- Users are concerned about the financial impact, suggesting the need for affordable solution.

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- When it comes to the user experience, it would be most useful for users to have a simple interface to avoid further stress while they're searching for their items in a highly emotional state.

In our next steps, we would use these findings to design and prototype major features of this Finding Service based off of the design implications above. From there, we would continue to improve upon the design by testing the different phases of the prototype with potential users and stakeholders.

Thank you. If you have questions about this report, please contact us at:

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Lost Item Finding Service

Final Report

HCI 445: User Research Methods

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Group Project Deliverable

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Executive Summary

This final report aims to explore the behavior of individuals when they are searching for lost items within their personal spaces. In total, eight participants were recruited through convenience sampling from social networks for the observational study, and eight more participants were recruited for the interview study through word of mouth and the CDM participant pool. With coding and affinity diagrams, many participants were discovered to misplace their daily essential items of phones, wallets, or keys. This result was supported by a 2004 study with similar discoveries. Both studies provided data which we analyzed for patterns of searching behaviors, emotional reactions, and techniques used by participants when trying to find a lost item. This ultimately led us to create two personas: Damien the dramatic and Carly the collected.

A technology based solution is introduced based on users preferences for design implications. In our study, there are several important findings from participants about finding misplaced items. systematic search strategies, their strong emotional reactions to the situation, and their process when seeking help from others.

In conclusion, we identified four themes for design implications: physical design, user experience, security, and cost. Users desire devices to be lightweight and have a slim form factor, keep their data secure, all at a reasonable price.

For future work, we recommend market research to pinpoint the range that users are willing to pay for such a system, as well as another round of interviews with a larger, more diverse, pool of participants.

Introduction

Misplacing items is a common and irritating phenomenon that happens to humans with no exceptions; it is an eventuality of our behaviors and environments. For this report, we conducted two studies: an observation report and a set of user interviews. Our report includes detailed findings and an examination of how individuals react when searching for a lost item. According to a 2017 study (Pertzov et al., 2017) on working memory, there's a connection between how one's most frequently used items may compete with each other in memory, resulting in errors when recalling their respective locations.

This report aims to provide a thorough analysis of the strategies, cognitive, and emotional patterns that help us understand the process individuals go through to find commonly lost items. We conducted observation sessions with eight participants who

were asked to follow a scenario in their personal space, where they mostly kept their items. The information gathered included patterns of searching behaviors, emotional reactions, and techniques they used. Comparing our findings to a similar study (Peters et al., 2004), we found that they had similar findings using a larger group sample. We also interviewed eight participants about their experiences with losing items as well as their experiences, if any, with using technology to both find lost items or prevent items from being lost in the first place. We also asked about things preventing them from using such systems and ideal features they would want from such a technology.

Method

Observation

Participants

We recruited eight participants through a convenience sampling of people in our social networks. We recruited people from our social network because we wanted to observe participants in their home environments, but weren't in a position to observe strangers in their homes in this study.

Participant ID	Ages	Gender	Occupation
1	37	Male	Software Engineering Manager
2	38	Male	UX Researcher
3	26	Female	Student
4	22	Female	Student
5	23	Female	Student
6	N/A	Female	Customer Service Representative
7	22	Female	Health Science undergraduate student
8	20	Male	Software Engineer

Data Collection Method

Each researcher recruited two participants and conducted individual observation sessions. The observation sessions took place inside the participant's home. We began by introducing ourselves and the project, and having the participant read and sign the informed consent form. Next, we asked some warm-up questions to get to know more

about each participant and items they commonly misplaced. Then, for the actual observational session, we asked them to do the following activity, while we recorded our observations:

Imagine you're about to leave your house for the day and you realize you're missing your [item from the warm-up question], please play out your next steps in trying to find that item.

After that, we asked any follow-up questions we had specific to our observations, as well as some predefined ones, and concluded the session.

Our full observation protocol is included in the appendix ([A1](#)).

Data Analysis Method

We collected all our observation notes using Optimal Workshop. We then created an affinity diagram to find common themes between our observations. The full affinity diagram can be found in the appendix ([A3](#)).

Interviews

Participants

We extended the study's participant pool beyond our personal social networks for the interview phase. Since we were not required to visit participants' houses as we were in the observation sessions, this method allowed us to include a wider spectrum of people. As part of our screening protocol, all participants had experience with misplacing items.

Participant ID	Ages	Gender	Occupation
1	24	Male	Sale man
2	21	Male	Student
3	28	Male	Unemployed components engineer
4	28	Male	Software Engineer
5	20	Male	Student
6	22	Female	Part time Medical Assistant
7	26	Female	Student
8	31	Male	Waiter

Data Collection Method

Using the data collected from the observation study, we created an interview protocol. Each researcher recruited two participants who passed the screening questions for the interview process. Interviews took place either online or in person, beginning with an introduction that provided an overview of the project and a consent form.

Following this, we started with some warm-up questions and then switched into deep, focused questions regarding their lost item experience, actively listening and engaging with the participants. Next, we discussed the potential use of technological solutions for finding lost items and features they would like to see in such technologies if possible.

Data Analysis Method

After collecting interview data, we created an inductive codebook based on the interview structure and answers. Subsequently, we coded all the interviews using Atlas.ti. Then, we made an affinity diagram and user spectrums. The full interview affinity diagram can be found in the appendix ([A6](#)). This affinity diagram and the user spectrums were used to create two personas, Damien and Carly. For each persona, we created a scenario and user journey map that illustrates how our future product could fit into their daily lives.

Findings

Observation Study Findings

Searching Methods

Participants were observed as having various searching methods during the observation and interview period. We organized the result from both into some common themes:

Having A Go-To Spot

We found that participants instinctively checked a usual spot relying on past experiences of finding lost items there. If the item wasn't found, they extended their search to various nearby locations, thoroughly examining places where small to medium-sized objects could be, such as under furniture, inside bags, or among personal effects. This two-pronged strategy reflects a pattern of starting with the most

likely location before transitioning to a more exhaustive search within the immediate vicinity.

Retracing Steps

Participants verbally and physically retraced their steps by repeating past actions related to the lost item.

Stressed Reactions

Many participants displayed strong emotional reactions when they were not able to find an item. Participant No.X “whined in a stressful way” to their partner when looking for said item in the activity. Participant No.Y was searching very quickly and frantically for their item, and participant No.Z was citing their finger anxiously while waiting for their mother to answer the phone.

Daily, But Essential Items

Most participants’ indicated that their most searched item, and item they can’t leave the house with, were items they interacted with the most such as water bottles, keys, smartphones, and wallets.

Asking For Help

All participants indicated that they would ask for assistance from a friend or family member if they couldn’t find an item.

Interview Findings

Impact of Losing Items

Emotional Impact

Majority of interviewees mentioned experiencing negative emotions such as panic, fear, frustration, stress, and embarrassment while searching for a misplaced item. One participant had a habit of losing their belongings, so they did not feel any resentment toward themselves or to the situation.

Relatedly, Most participants expressed feeling relieved immediately after they found their missing item.

Lost Time

About half of the participants mentioned finding their misplaced items in less than an hour, but a few spent either hours or days.

Daily, But Essential Items

When answering questions in the interview, all eight participants said that the items they lost the most frequently were daily items such as phones, wallet/keys, and personal accessories. All participants described misplacing their items in areas of their home as opposed to any other daily location, such as their car, school, or workplace.

Preventative Strategies and Search Methods

Participants mentioned that they either had a specific organizational system, took precautionary measures like reminders or notes, or simply attempted to remember where their items were.

The interviews corroborated the search strategies we identified during our observational study. The majority of the participants mentioned that their first reaction was to check the last place they thought the item was used, then search every location that was remotely connected to the item. Also, several participants asked for help from their friends or families.

Reason's Items Were Misplaced

The majority of the participants stated that they merely placed their items at uncommon locations. A few stated that they were either distracted or rushing when they were placing their item somewhere.

Desired Technological Features

During our interviews, we found clear commonalities between features of a finding service that users desire. The majority of the participants would love that their ideal solution had precise and reliable tracking properties. Another important category of features was design, with half of the participants noting that a small form factor was important to them, and two participants noting that overall aesthetics was a concern.

The most repeated concern, or barrier to entry, for such a service was about the safety and security of the service. One participant even remarked that they'd be worried that a

bad actor would be able to tell that his backpack would be a good target for theft if they were able to tell it contained multiple item tracking devices.

Personas

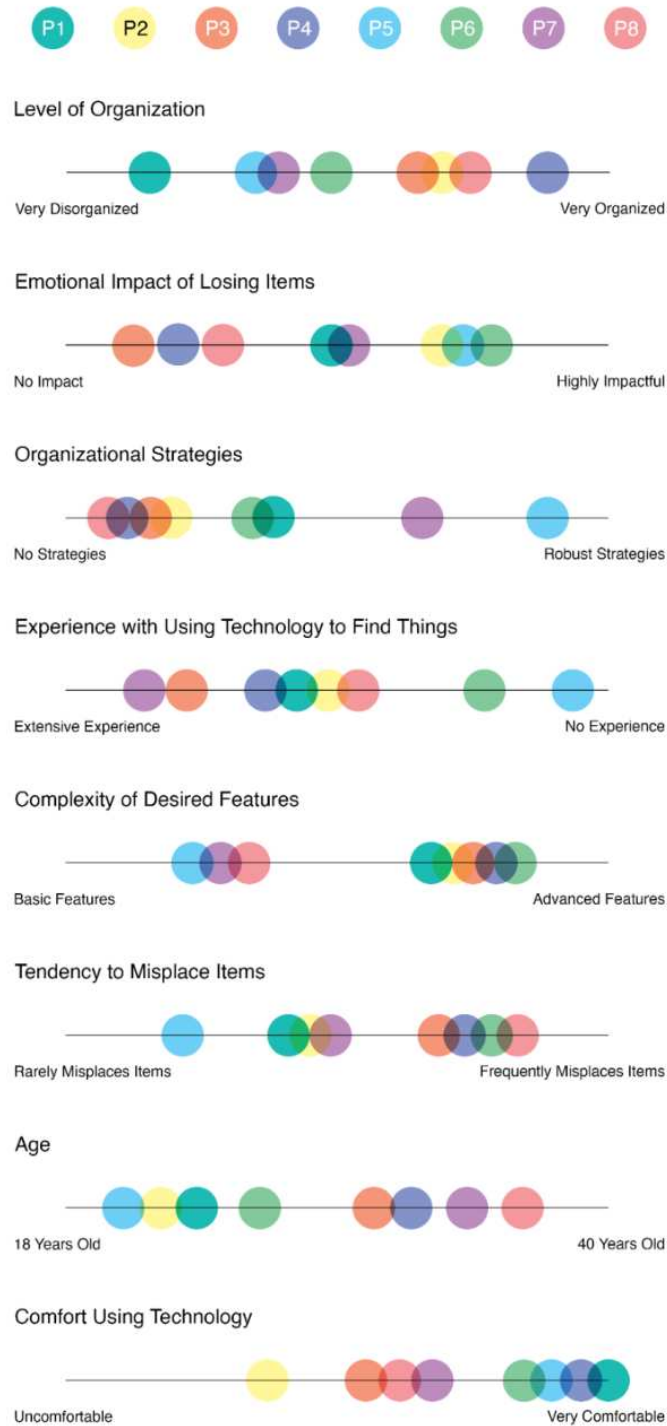
Spectrum Analysis

We took our interview participants and placed them on eight spectrums, using common themes from their interviews.

First we created spectrums related to demographic information we felt was most salient to our service: comfort using new technology and age. Next, we focused on attributes that might impact how frequently a user would use our service: level of organization, organizational strategies, and tendency to misplace items. Then, we created spectrums that related to how they would want to interface with our service: experience with using "finding" technologies, complexity of desired features, and emotional impact of losing items.

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Persona 1: Damien the Dramatic

Name: Damien

Pronouns: They/them

Age: 25

Occupation: Graduate Student

Bio

Damien is a relatively organized person, with certain areas of their living space kept tidy (like their bookshelf) while others can tend to get messier (their desk). They sometimes misplace items when rushed or distracted. Damien has lost meaningful sentimental items before like a gifted necklace, causing significant distress until the item was recovered. Part of their organizational system includes trying to use systems like post-it notes and designated spots to keep track of daily items like keys, wallet, and phone. Because Damien doesn't lose items very frequently, when they do lose items, they have a higher emotional reaction until they're to find what's missing.

Pain Points:

- Loses items due to being in a rush, being distracted, and/or accidentally placing things in unusual spots
- Experiences high levels of panic, frustration, and stress when items go missing, especially when it's a sentimental item
- Current organizational systems like notes and set places are not foolproof
- Losing items has caused inconveniences like having to replace IDs/cards and update payment info

Needs:

- A simple, hassle-free way to quickly locate misplaced items
- A solution that works seamlessly even when stressed or rushed
- Long range tracking ability
- No bulky accessories that add too much weight to everyday carry items
- Affordable pricing
- Ability to locate items even without internet/bluetooth connection
- In terms of an object finding solution, Damien would likely appreciate something that can discreetly track commonly misplaced everyday items like keys, wallet, phone without added bulk. The solution should have a simple user interface that makes locating the item easy even when frustrated. Long range tracking, offline usability, and affordable pricing are also priorities. Any emotional reassurance

like soothing sounds during the search could also provide value during stressful lost item scenarios.

Persona 2: Carly the Collected

Name: Carly

Pronouns: She/her

Age: 23

Occupation: Undergraduate Student

Bio

Carly is a naturally untidy person, yet she has her own ways of organizing or placing her items to keep them close and easy to locate. Her emotional reaction to losing an item is lower; she views them as minor hassles rather than significant losses. This may be because she misplaces items frequently and she has become accustomed to the experience. She is not as fluent with technology and would prefer to use simple features over complex ones. However, Carly's openness to new technologies shows that she is willing to try solutions that will improve her everyday routine.

Pain Points:

- Carly's main pain point is her own natural disorganization, which makes it difficult for her to track and find her personal items.
- Carly is unable to benefit from the technological solution because of her resistance or ignorance of using those solutions, which show a need for more intuitive and user-friendly design.

Needs:

- A better way to be more organized.
- Items that are simple to put into her daily life and improve her ability to organize possessions without significantly changing her daily routine.
- Adaptable technology with basic friendly features to help her find lost items so she can get a better experience and be more comfortable with it.
- Accessible technology to give her more options.

Scenarios

Damien

As a busy graduate student, Damien realizes they really struggle with the stress they experience when something important of theirs goes missing. They especially don't have the time to search for something when it's time for them to leave for class or important meetings.

Damien's current organizational system is to make mental notes of where they leave things, or to consistently return their daily items to known, set locations. However, this system falls apart when they absent-mindedly set something in an unusual place and forget to make a note of it.

Damien comes across our Item Finding Service and, because Damien is very comfortable with new technology, immediately configures advanced settings and reminders upon returning to their home after purchasing.

The next day, Damien is about to leave for class when they realize they can't find their USB drive that has their presentation on it. Instead of panicking, Damien pulls up our Item Finding Service and searches for the USB drive. The system tells Damien that the USB drive is on the kitchen table and Damien is able to quickly find it. They leave for the university with the presentation safely in hand, feeling happy and grateful of how technology helped them transform what could have been a trying morning into a successful start to the day.

Carly

Carly is a disorganized undergraduate student, but for the most part, her system works for her. She realizes she spends a bit of time looking for things before she leaves her dorm room, despite generally knowing where things are. In an attempt to be late less often, Carly decides to buy our Item Finding Service and, as a not very tech-savvy person, completes the basic set-up flow before continuing on with her day.

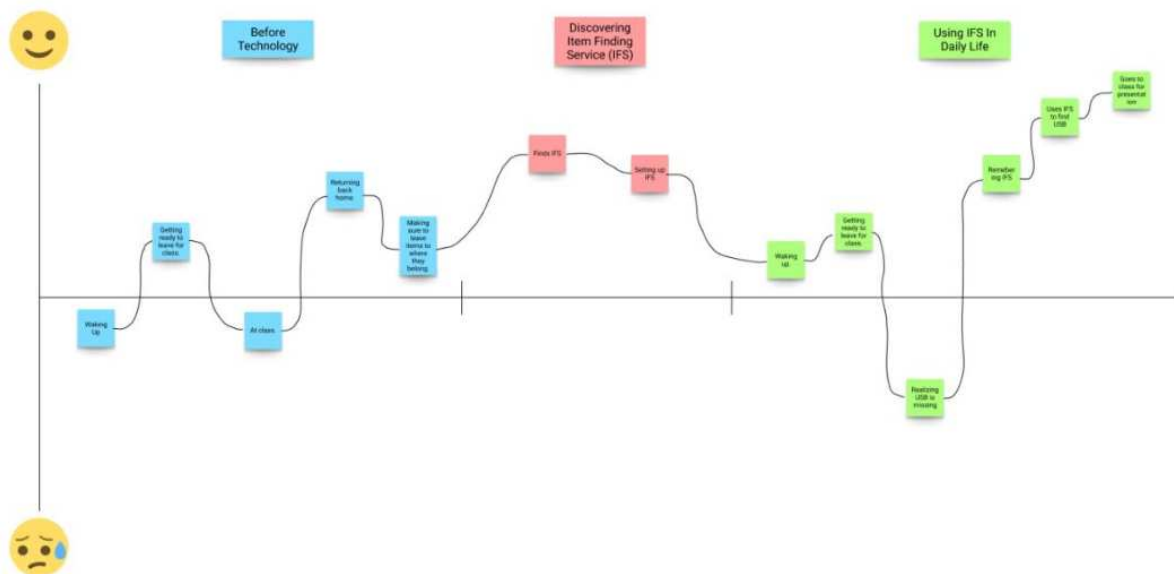
As Carly is about to leave her dorm to get dinner with some friends, she realizes that she's lost her wallet, yet again. Frustrated by this delay, she opens up the Item Finding Service app and clicks "Wallet." The app points Carly in the direction of her wallet while the hardware in her wallet makes a faint noise, allowing her to pinpoint it immediately. Carly's able to be out the door in no time and, to the surprise of her friends, is the first person to arrive at the restaurant.

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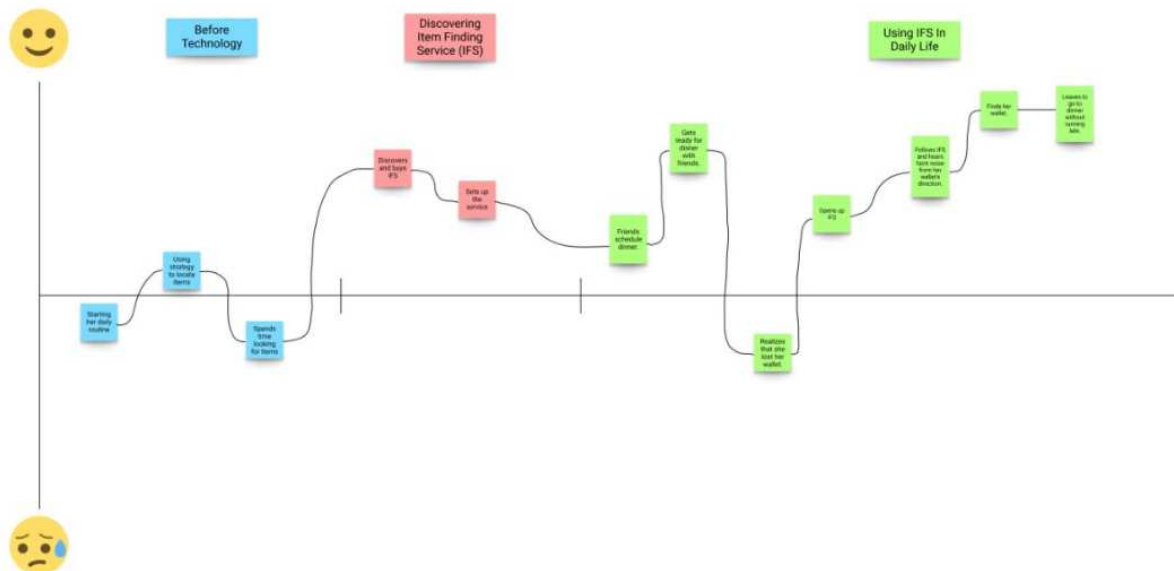
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User Journey

Persona: Damien the Dramatic



Persona: Carly the Collected



Priority Matrix

During our discussion, we analyzed each feature's possible impact and feasibility in order to determine design implications. This analysis helped us define priorities for the development of design concepts. We organized these into the priority matrix below. (see Table 2: Priority Matrix)

Table 2: Priority Matrix

No.	Feature	Category	Priority	Impact	Feasibility
1.	Compact Design	Design	High	High	High
2.	Working Long-Range	Functionality	High	High	Medium
3.	Offline Functionality	Functionality	Medium	High	Medium
4.	Simple Interface	User Experience	High	High	High
5.	Privacy and Security	Security	High	High	Low
6.	Affordability	Cost	Medium	High	Medium
7.	Versatile Use	User Experience	Medium	Medium	High

Discussions

Our study started with the following research question: What are user needs for the design of a technology aimed to assist users in finding commonly lost or misplaced possessions? From there we set out to understand behaviors and emotions during the search for lost items at home.

In our observations, we found that participants typically used a systematic search strategy, starting from common locations, retracing steps, and then expanding the search if needed. These methods were also observed in Peters' study and that participants in his study tend to misplace frequently used items (Peters et al., 2004). This reflects both logical tactics and the cognitive aspects of memory and spatial orientation.

Between our observational study and our interviews we identified four clear themes when it comes to implications for design: physical design, user experience, security, and cost.

Physical Design

In our interviews, users expressed the importance of the physical design in several key ways. Firstly, users noted the importance of slim profiles and lightweightness of any physical elements of this system, especially if anything is going to be adhered or attached to their items. One interviewee also noted how, in the case of any potential use of adhesives, they would have concerns over potentially damaging the item they were attaching something to.

User Experience

From our observational studies, we found that stress and anxiety play a significant role in the search process, impacting search efficiency. This highlights the need for item-tracking solutions that support users not just functionally but also emotionally. These findings suggest a need for intuitive and empathetic technological aids. Future designs should integrate technical effectiveness with an understanding of human emotions, creating user-centered products that alleviate stress and enhance user experience.

When it comes to the user experience, we also found that our participants' search strategies were inherently spatial, which is corroborated by the 2004 Peters et al. study. The design implication is that the user interface should reflect that in some way and take advantage of the spatial thinking that users' exhibit while searching for lost items.

Based on the interviews, other important considerations for the user experience include long-range and offline functionality, as well as a simplified user interface. While some users may want the option for more advanced settings, a purposefully simple user interface could also tie into the previously mentioned notion to support users emotionally while they're searching for items.

Security

From our interviews it became clear that security was a large concern for users. This often came up when participants were asked about their concerns with potentially using technology in order to keep track of and find their items. This makes sense as technology isn't perfect and we're talking about technology that could potentially track a user's location, or at the very least the location of their valuable items.

Cost

While only two interviewees mentioned cost as an obstacle or concern when it comes to this technology, we wanted to call it out in this early stage of the design process in order

to properly address it. While ideating on potential design solutions and prototypes, we should be mindful of the end-user cost, especially because few of the participants interviewed expressed significant concern over the financial impact losing items has had on their lives. This implies that paying a large amount of money for a system to prevent themselves from losing things in the future may not make economical sense for most users. Market research in this area might be needed in order to pinpoint the exact range potential users would be willing to pay for this service.

Limitations and Future Work

One major limitation we came across during our observation study was that this was a contrived activity as we could not manufacture a situation in which a participant was actually searching for a lost item, especially in a way that simulates the potential stress and immediacy of a real life situation. Secondly, in order for our observation protocol to work, we needed access to the participants' homes, and as student researchers with limited time to schedule sessions and find participants, this resulted in us opting to use friends and acquaintances as our participants. Not only was this because of the limited time, but we wouldn't necessarily be able to schedule sessions in pairs, and we decided we weren't comfortable doing observational studies alone in strangers' homes.

For our interview study, the demographics of our participants skewed younger and potentially as a result, more comfortable with technology than the general population. This is most likely due to us recruiting participants using word of mouth and the CDM Participant Pool, yielding us participants similar to our age and more likely to be students or young professionals themselves.

To get a better sense of other demographics' needs when it comes to a technological solution for finding lost items, we would recommend that future work involves another interview study. This interview study should have a larger number of participants, with a strong focus on recruiting diverse participants that more closely reflects the general population.

After that, the next steps would be to start formally designing and prototyping the Item Finding Service. User testing and feedback at the varying stages of prototyping would allow us to better iterate on the design, and create a more useful product.

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Appendix

A1: Observation Protocol

Step 1: Introduction

Introduce Yourself and the Project:

Hello, my name is [Your Name], and I'm conducting a study on how individuals manage and locate personal items they tend to misplace.

Purpose of the Study:

The purpose of this study is to better understand the challenges and strategies people use to track their items. Only the team will know your name and a fake name will represent your responses when we present our findings in class.

Procedure Overview:

As we go through the tasks I may ask you some questions. Note there aren't any right or wrong answers, or right or wrong ways to perform the tasks. My goal is to observe you and not to judge you in any way. The study will be recorded, to refer back to it later and for a seamless and quick experience. I might ask small questions for clarification and take some side notes on the side to refer back to later.

Informed Consent:

To continue, I will need your informed consent by having you read our Informed Consent Form. In essence, this form details the purpose of this study, details on how your data will be kept confidential, and contact information should you have any questions or concerns. Keep in mind that your participation is completely voluntary and you may leave this study, at any time.

Provide the informed consent form.

Please sign in here.

Now that you have signed the consent form, we can start with some warm-up questions.

Step 2: Warm-Up

I will start asking some warm up questions now to get the ball rolling. You can take as much time as you need and are allowed to exit at any time.

1. Tell me a little about yourself?
 - a. Like your name, hobbies, occupation, interests, education, age, daily routine.
2. What are the most commonly misplaced items in your daily life, and do you notice any patterns in how and where you misplace them?
3. What is the item, or items, that you never leave the house without?

Step 3: Observation Scenario

We will begin the observation scenario now. You can take as much time as you need and are allowed to exit at any time. While you're doing the activity, please say your thought process out loud.

- Imagine you're about to leave your house for the day and you realize you're missing your [item from the warm-up question], please play out your next steps in trying to find that item.

Step 4: Wrap-Up

Follow-Up Questions:

1. [Any follow-up or clarifying questions specific to this participant's session]
2. If you weren't alone, would you involve others, like family or friends, in helping you locate misplaced items? How do they contribute?
3. Were there any strategies you used today that you find particularly effective or ineffective in your daily life?
4. Are there specific strategies or tools you use to help you remember where you placed certain items?

Conclusion:

- Thank you [participants name] for participating in our study, you've been a great help! Are there any questions you have for me before we finish?

A2: Observation Informed Consent Form

FINDING LOST OBJECTS: OBSERVATION STUDY

Principal Investigators: The graduate students Suleyman Sarigoz, Michael Turenne, Tee Trakulsuebchai, Abdal Jarwan at the HCI Jarvis School of Computing and Digital Media, DePaul University

Institution: DePaul University, Chicago, Illinois, USA

School: DePaul University

College: Jarvis School of Computing and Digital Media

Faculty Advisor: Oliver Alonzo, HCI Jarvis School of Computing and Digital Media, DePaul University

Key Information:

What is the purpose of this research?

We are asking you to be in a research study because we are trying to learn more about how individuals manage and locate personal items they tend to misplace. This study is being conducted by Suleyman Sarigoz, Michael Turenne, Tee Trakulsuebchai, Abdal Jarwan, graduate students at DePaul University as a requirement to obtain their master's degree. This research is being supervised by their faculty advisor, Oliver Alonzo.

There may be other people on the research team assisting with the study.

We hope to include about 8 people in the research.

Why are you being asked to be in the research?

You are invited to participate in this study because you have had experience misplacing personal items.

You were chosen for this study because you meet the inclusion criteria of having experienced misplacing an item for more than 5 minutes recently, and are proficient in speaking English. You must also be 18 or older to be in this study. This study is not approved for the enrollment of people under the age of 18.

What is involved in being in the research study?

If you agree to be in this study, being in the research involves:

- Completing an observational session simulating how you would deal with a situation where you have misplaced an item.
- Answering a survey to collect general information about you, such as gender, occupation, age, the type of items you misplace.

The interview will be audio recorded and transcribed into written notes later in order to get an accurate record of what you said.

Are there any risks involved in participating in this study?

You may feel uncomfortable, embarrassed, sad, or angry about answering questions or performing the tasks for the observational study. You do not have to answer any question you do not want to and are free to stop the observational session at any time.

There is the possibility that others in the team will find out what you have said, but you will be given a fake name when analyzing the data to prevent anyone from connecting your responses to you.

Are there any benefits to participating in this study?

You may benefit from the research by sharing insights that may one day become a product that aids in individuals managing and locating misplaced personal items. Beyond that, you will not personally benefit from being in this study.

We hope that what we learn will help other people in the future by creating a better tool to manage misplaced personal items, and contribute to the knowledge in the field of memory.

Other Important Information about Research Participation

Can you decide not to participate?

Your participation is voluntary, which means you can choose not to participate. There will be no negative consequences, penalties, or loss of benefits if you decide not to participate or change your mind later and withdraw from the research after you begin participating.

The researcher may remove you from the study without your consent, if you do not follow the instructions, if your situation changes and you no longer meet the inclusion criteria for the study, or if you are no longer able to complete the study tasks or come for study visits.

Who will see my study information and how will the confidentiality of the information collected for the research be protected?

The research records will be kept and stored securely. Your information will be combined with information from other people taking part in the study. When we write about the study or publish a paper to share the research with other researchers, we will write about the combined information we have gathered. We will not include your name or any information that will directly identify you. Some people might review or copy our records in order to make sure we are following the required rules, laws, and regulations.

To prevent others from accessing our records or identifying you, should they gain access to our records, we have put some protections in place. These protections include using a code (a fake name, a study ID number, etc.) for you and other people in the study and keeping the records in a safe and secure place using a password-protected Google Drive folder. The audio recordings will be kept until accurate written notes have been made, then they will be destroyed.

You should know that there are some circumstances in which we may have to show your information to other people. For example, the law may require us to show your information to a court or to tell authorities if you report information about a child being abused or neglected or if you pose a danger to yourself or someone else.

Who should be contacted for more information about the research?

Before you decide whether to accept this invitation to take part in the study, please ask any questions that might come to mind now. Later, if you have questions, suggestions, concerns, or complaints about the study or you want to get additional information or provide input about this research, you can contact any of the researchers using the contact information listed here:

- Michael Turenne at (414) 841-6365, or mturenn1@depaul.edu.
- Suleyman Sarigoz at (408) 429-4979 or ssarigoz@depaul.edu
- Teetawat Trakulsuebchai (224) 384-9888 or ttrakuls@depaul.edu
- Abdalrahman Jarwan (347) 827-5097 or ajarwan@depaul.edu

You can also contact the faculty sponsor, Oliver Alonzo, at oliver.alonzo@depaul.edu.

You may also contact DePaul's Office of Research Services if:

- Your questions, concerns, or complaints are not being answered by the research team.
- You cannot reach the research team.
- You want to talk to someone besides the research team.

LOST ITEM FINDING SERVICE

Jarwan, Sarigoz, Trakulsuebchai, Turene

You will be given a copy of this information to keep for your records.

Statement of Consent from the Subject:

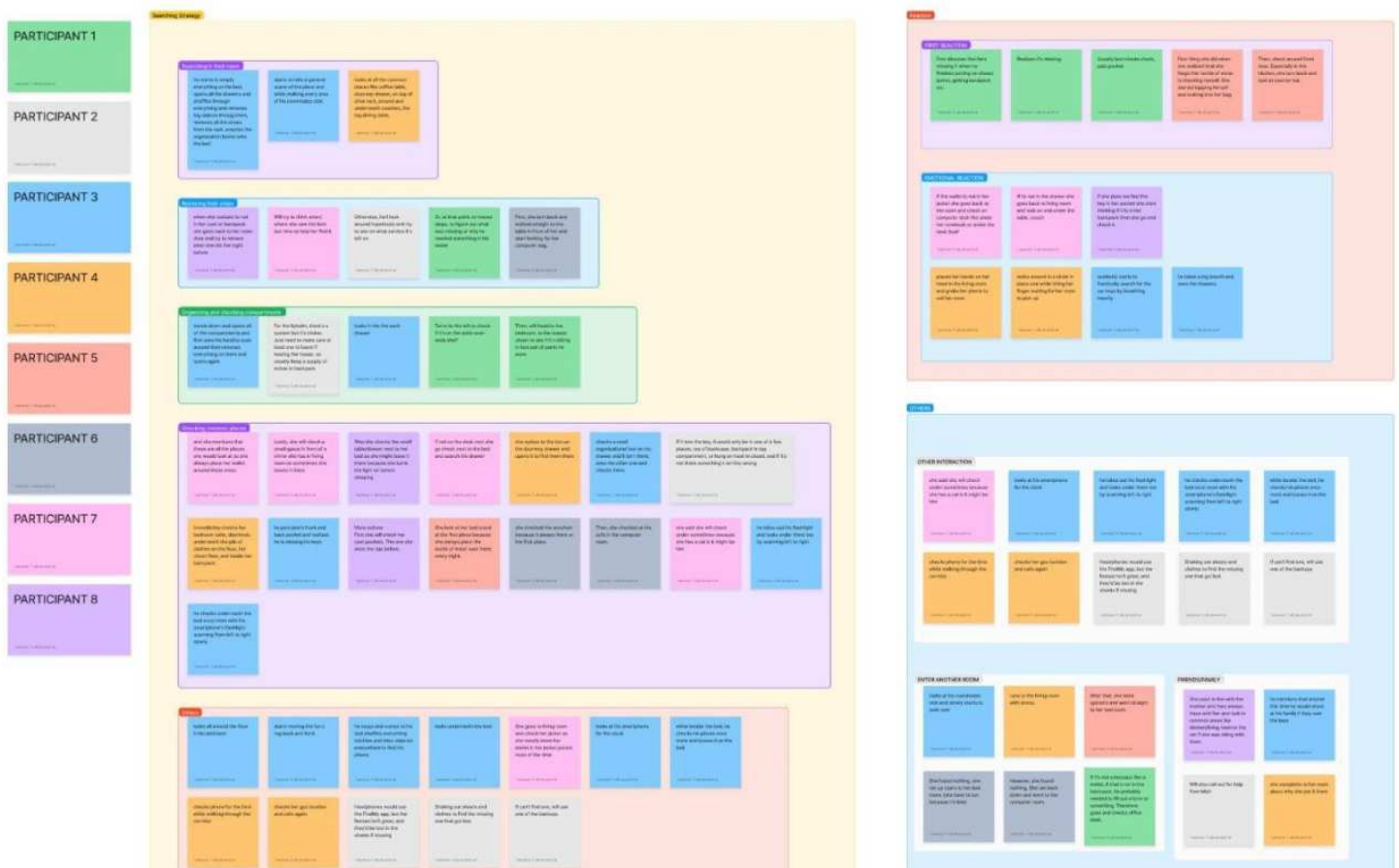
I have read the above information. I have had all my questions and concerns answered. By signing below, I indicate my consent to be in the research. By completing the interview you are indicating your agreement to be in the research.

Signature: _____

Printed name: _____

Date: _____

A3: Observation Affinity Diagram



A4: Interview Informed Consent Form

FINDING LOST OBJECTS: INTERVIEW

Principal Investigators: The graduate students Suleyman Sarigoz, Michael Turenne, Tee Trakulsuebchai, Abdal Jarwan at the HCI Jarvis School of Computing and Digital Media, DePaul University

Institution: DePaul University, Chicago, Illinois, USA

School: DePaul University

College: Jarvis School of Computing and Digital Media

Faculty Advisor: Oliver Alonzo, HCI Jarvis School of Computing and Digital Media, DePaul University

Key Information:

What is the purpose of this research?

We are asking you to be in a research study because we are trying to learn more about how individuals manage and locate personal items they tend to misplace. This study is being conducted by Suleyman Sarigoz, Michael Turenne, Tee Trakulsuebchai, Abdal Jarwan, graduate students at DePaul University as a requirement to obtain their master's degree. This research is being supervised by their faculty advisor, Oliver Alonzo.

There may be other people on the research team assisting with the study.

We hope to include about 8 people in the research.

Why are you being asked to be in the research?

You are invited to participate in this study because you have had experience misplacing personal items.

You were chosen for this study because you meet the inclusion criteria of having experienced misplacing an item for more than 5 minutes recently, and are proficient in speaking English. You must also be 18 or older to be in this study. This study is not approved for the enrollment of people under the age of 18.

What is involved in being in the research study?

If you agree to be in this study, being in the research involves:

- Completing an interview session.
- Answering a survey to collect general information about you, such as gender, occupation, age, the type of items you misplace.

The interview will be audio recorded and transcribed into written notes later in order to get an accurate record of what you said.

Are there any risks involved in participating in this study?

You may feel uncomfortable, embarrassed, sad, or angry about answering questions in the interview. You do not have to answer any question you do not want to and are free to stop the interview session at any time.

There is the possibility that others in the team will find out what you have said, but you will be given a fake name when analyzing the data to prevent anyone from connecting your responses to you.

Are there any benefits to participating in this study?

You may benefit from the research by sharing insights that may one day become a product that aids in individuals managing and locating misplaced personal items. Beyond that, you will not personally benefit from being in this study.

We hope that what we learn will help other people in the future by creating a better tool to manage misplaced personal items, and contribute to the knowledge in the field of memory.

Other Important Information about Research Participation

Can you decide not to participate?

Your participation is voluntary, which means you can choose not to participate. There will be no negative consequences, penalties, or loss of benefits if you decide not to participate or change your mind later and withdraw from the research after you begin participating.

The researcher may remove you from the study without your consent, if you do not follow the instructions, if your situation changes and you no longer meet the inclusion criteria for the study, or if you are no longer able to complete the study tasks or come for study visits.

Who will see my study information and how will the confidentiality of the information collected for the research be protected?

The research records will be kept and stored securely. Your information will be combined with information from other people taking part in the study. When we write about the study or publish a paper to share the research with other researchers, we will write about the combined information we have gathered. We will not include your name or any information that will directly identify you. Some people might review or copy our records in order to make sure we are following the required rules, laws, and regulations.

To prevent others from accessing our records or identifying you, should they gain access to our records, we have put some protections in place. These protections include using a code (a fake name, a study ID number, etc.) for you and other people in the study and keeping the records in a safe and secure place using a password-protected Google Drive folder. The audio recordings will be kept until accurate written notes have been made, then they will be destroyed.

You should know that there are some circumstances in which we may have to show your information to other people. For example, the law may require us to show your information to a court or to tell authorities if you report information about a child being abused or neglected or if you pose a danger to yourself or someone else.

LOST ITEM FINDING SERVICE

Jarwan, Sarigoz, Trakulsuebchai, Turene

Who should be contacted for more information about the research?

Before you decide whether to accept this invitation to take part in the study, please ask any questions that might come to mind now. Later, if you have questions, suggestions, concerns, or complaints about the study or you want to get additional information or provide input about this research, you can contact any of the researchers using the contact information listed here:

- Michael Turenne at (414) 841-6365, or mturenn1@depaul.edu.
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You will be given a copy of this information to keep for your records.

Statement of Consent from the Subject:

I have read the above information. I have had all my questions and concerns answered. By signing below, I indicate my consent to be in the research. By completing the interview you are indicating your agreement to be in the research.

Signature: _____

Printed name: _____

Date: _____

A5: Interview Protocol

Research Question

What are user needs for the design of a technology aimed to assist users in finding commonly lost or misplaced possessions?

Screeners Questions

1. Are you 18 years old or older? [Exclusionary answer: NO]
2. Are you proficient in speaking English? [Exclusionary answer: NO]
3. Have you experienced misplacing an item for more than 5 minutes recently?
[Exclusionary answer: NO]

Introduction

Introduce Yourself and the Project:

Hello, my name is [your name], and my team is conducting interviews on how individuals manage and locate personal items they tend to misplace.

There are no right or wrong answers. Today's interview session should take approximately 30 minutes without interruptions. Please refrain from using your cell phone during the interview.

Purpose of the Study:

The purpose of this study is to better understand the challenges and strategies people use to track their items. Only the team will know your name and a fake name will represent your responses when we present our findings in class.

Procedure Overview:

As we go through the interview questions I may ask you some clarifying questions. Again, please note that there aren't any right or wrong answers to these questions. My goal is to simply collect your answers and not to judge you in any way. The interview will be recorded, to refer back to it later and later transcribed for analysis. I might ask small questions for clarification and take some side notes on the side to refer back to later.

Informed Consent:

To continue, I will need your informed consent by having you read our Informed Consent Form. In essence, this form details the purpose of this study, details on how your data will be kept confidential, and contact information should you have any questions or concerns. Keep in mind that your participation is completely voluntary and you may leave this study, at any time.

*****Provide the informed consent form.*****

Please sign in here.

Now that you have signed the consent form, we can start with some warm-up questions.

Warm Up Questions:

1. Please tell me a little bit about yourself.
2. In your house, where would you describe as the messiest?
 - a. The most organized?
3. Are there any specific habits or rituals you've developed to prevent misplacing items?

Deep Focus

A. Lost Item Experience

1. Can you tell me about the last time you lost or misplaced an item?
 - a. *Follow-up questions to get more information, depending on answer:*
 - i. *How long was it lost for?*
 - ii. *What was the experience like emotionally?*
 - iii. *How did you end up finding it?*
2. Can you tell me a time that you have lost a very meaningful object?
 - a. *Follow-up questions to get more information, depending on answer:*
 - i. *How long was it lost for?*
 - ii. *What was the experience like emotionally?*
 - iii. *Please explain any side effects that have arisen because of the lost object.*
 - iv. *How did you end up finding it?*
3. Can you think of common themes, if any, to how you tend to lose items?
4. Can you tell me about the financial or emotional impacts, if any, losing things has had on your life?

B. Technological Solutions

1. What's your opinion on using technology to find lost objects?
2. Have you ever used a technological solution for finding a lost item?
 - a. If yes, can you tell me about the experience of using it?
 - b. If not, are there any barriers or concerns you have with using such a system in the future?

Retrospective

1. In a perfect world, if there was a technology that could improve finding an item you've lost, what kind of features would that technology have?
2. How could a technological solution be useful and enjoyable to use even if you're stressed while looking for a lost item?

Demographics

1. What is your age?
2. What is your gender?
3. What is the highest level of education you have completed?
4. What is your occupation?
5. On a scale of 1-5, 5 being the highest, how would you rate your comfort with using new technology?

Closing

Thank you [participants name] for participating in our study, you've been a great help!
Are there any questions you have for me before we finish?

A6: Interview Affinity Diagram



A7: Interview Codebook

A. Demographics

Code under: **Demographics**

Attribute codes to use when the participant is answering demographic questions, or telling the interviewer about themselves.

A.1. Age

Code under: **Demographics_Age**

Code when the participant is describing their age

A.2. Occupation

Code under: **Demographics_Occupation**

Code when the participant is describing their occupation.

A.3. Education

Code under: **Demographics_Education**

Code when the participant is describing their highest level of education.

A.4. New Technology Comfort

Code under: **Demographics_New Technology Comfort**

Code when the participant is describing their comfort with using new technologies.

A.5. Gender

Code under: **Demographics_Gender**

Code when the participant is describing some other aspect of demographic information not explicitly asked for.

A.6. Other [Description]

Code under: **Demographics_Other_[Description]**

Code when the participant is describing some other aspect of demographic information not explicitly asked for.

B. Living Environment

Code under: Living Environment

B.1. Location

Code under: **Living Environment_Location**

Code when participants are describing a specific location in their living environment.

B.2. Organized

Code under: **Living Environment_Organized**

Code when participants are describing particularly organized parts of their living environment.

B.3. Messy

Code under: **Living Environment_Messy**

Code when participants are describing particularly unorganized or messy parts of their living environment.

C. Preventative Strategies

Code under: **Preventative Strategies**

Code when the participant is discussing any strategies they may use to prevent losing items in the first place.

C.1. Location [Description]

Code under: **Preventative Strategies_Location_[Description]**

Code when describing a specific location used in preventative strategies.

C.2. Tool [Description]

Code under: **Preventative Strategies_Tool_[Description]**

Code when describing a specific location used in preventative strategies.

C.3. Mental Model [Description]

Code under: **Preventative Strategies_Mental Model_[Description]**

Code when participants are discussing a mental mode, mnemonic device, or other cognitive tool to prevent themselves from losing items.

D. Lost Item

Code under: **Lost Item**

Code for documenting the result or impact of losing an item.

D.1. Item [Description]

Code under: **Lost Item_Item_[Description]**

Code to use when the participant is describing the actual item that they have lost.

D.2. Impact

Code under: **Lost Item_Impact**

Code category to use for the following elements describing the impact on the participant of losing an item.

D.2.1. Emotion

Code under: **Lost Item_Impact_Emotion**

Code when describing specific emotions that arise as a result of losing, or searching for, an item.

D.2.2. Financial

Code under: **Lost Item_Impact_Financial**

Code when the participant describes any financial impacts from losing an item.

D.2.3. Time

Code under: **Lost Item_Impact_Time**

Code to use when the participant is describing how much time they spent finding a lost item and/or how much time they may have spent replacing or recreating a lost item.

D.3. Reason

Code under: **Lost Item_Reason**

Code to use for any underlying reason or reasons that an item became lost in the first place.

E. Search Strategy

Code under: **Search Strategy**

Code to use when discussing strategies for finding a lost item

E.1. Success

Code under: **Search Strategy_Success**

Code when participants are describing a search strategy that was used to successfully find an object.

E.1.1. Location [Description]

Code under: **Search Strategy_Success_Location_[Description]**

Code when describing the specific place where an item was found.

E.2. Failure

Code under: **Search Strategy_Failure**

Code when participants are describing a search strategy that was not used to successfully find an object.

F. Technology

Code under: **Technology**

Code used when the participant is discussing any type of technological interventions for finding a lost object, or the prevention of losing a lost object.

Note: If this overlaps with another coding category, such as **Search Strategy** or **Preventative Strategies**, code both.

F.1. Design Implications

Code under: **Technology_Design Implications**

Code for when a participant mentions design features in existing or hypothetical technology.

F.1.1. Idealized

Code under: **Technology_Design Implications_Idealized**

Code when mentioned an ideal or desired design for a technology based solution to finding lost items, or the prevention of losing items.

F.1.2. Existing (Desired)

Code under: **Technology_Design Implications_Existing_Desired**

Code when an existing technological solution's design is mentioned in a positive light, or as a desired feature for other solutions.

F.1.3. Existing (Undesired)

Code under: **Technology_Design Implications_Existing_Undesired**

Code when an existing technological solution's design is mentioned in a negative light, or as an undesired feature for other solutions.

F.2. Emotion

Code under: **Technology_Emotion**

Code for when a participant is discussing how they feel about using technological interventions, or have felt while using technological interventions.

F.3. Obstacles

Code under: **Technology_Obstacles**

Code for when a participant is discussing any obstacles or barriers they've encountered when thinking about using technological interventions for finding lost objects, or the prevention of losing objects in the first place.